RESEARCH REPORT SERIES
(Survey Methodology #2024-05)

Usability of Quick Response (QR) Code as a Method to Access a Survey Using a Smartphone

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Report issued: August 2024

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Abstract

Logging into a survey can be burdensome, particularly when using the small keyboard in a smartphone to type in the survey's webpage. In this study, we explored the usability (effectiveness, efficiency, and satisfaction) of Quick Response (QR) codes as a method to access an online survey through a smartphone. Twenty respondents (average age: 29.9 years) were presented with instructions to scan a QR code to access a survey in their smartphone. All respondents successfully scanned the QR code without any usability issues, in an average time-on-task of 12.4 seconds and rated the process as "extremely easy." One limitation of this study is that we were unable to find respondents without previous QR code experience. This, however, may provide evidence for the prevalence of QR codes in everyday activities, and may be good news for survey methodologists looking for a way to reduce respondent burden when accessing a survey through a smartphone.

Keywords: Mobile survey, Alternative login, QR Code, Usability

Suggested Citation: Alda Rivas, Ashley Peterson, Anthony Schulzetenbers and Lin Wang. (**2024**). **Usability of Quick Response (QR) Code as a Method to Access a Survey Using a Smartphone.** *Research and Methodology Directorate, Center for Behavioral Science Methods Research Report Series (Survey Methodology #2024-05).* U.S. Census Bureau. Available online at <https://www.census.gov/library/working-papers/2024/adrm/rsm2024-05.html>

1. Introduction

In this modern era, many organizations conduct surveys online, directing respondents to their surveys through emails, websites, or other online platforms. This trend may be due to the fact that 93% of adults in the United States use the internet (Pew Research Center 2021). When the invitation to a survey is sent through an email or website, respondents can click on a link to easily access online surveys without having to manually type the web address (Bethlehem and Biffignandi 2021). Organizations may also send a paper invitation to respondents through the mail. This option results in higher response rates compared to email (Sakshaug, Vicari, and Couper 2019), but it also has some potential disadvantages.

When accessing an online survey from a paper invitation, respondents complete the process of opening a browser and manually entering the web address that is included in the invitation (Sakshaug et al. 2019). The web address may include alphanumeric characters, which increases the possibility of respondent error during manual entry. This increase in burden for potential respondents may lead to lower response rates (Galesic 2006), which in turn, leads to an increase in total survey error (Biemer 2010). Considering the number of people in the United States who own a smartphone (85% as of 2021; Pew Research Center 2021), and the increasing number of respondents who access online surveys on their smartphones (Antoun, Couper, and Conrad 2017), improving the process to access surveys through mobile devices may have a positive impact on response rates. One potential avenue to reduce respondent burden is to allow respondents to access the online survey through a Quick Response (QR) code.

QR codes are machine-readable optical labels that contain pre-programed information, often a website link, which is encoded within its matrix. Figure 1 provides a visualization of a QR code.



Figure 1. An example of a QR code that contains the link to a website.

To access an online page using a QR code, the built-in camera on a smartphone is used to scan the QR code. After placing the QR code in the camera's focus, a notification appears on the screen displaying the URL that is embedded (although older phones may require installation of a third-party application). When the notification is tapped, the browser opens the website that is linked to the QR code.

Several studies examining the use of QR codes to access surveys have documented increased survey response rates (Harrison et al. 2019; Masih 2022; Ramalingam, Tran, and

Gangopadhyay 2020). Furthermore, QR codes have been perceived as more usable than other data entry forms, such as manually typing in a personalized code (Marky et al. 2019), or filling out a paper form (Snyder et al. 2018). However, other studies exploring the relationship between QR codes and survey response rates have not reached such positive conclusions.

Marlar (2018) conducted a study that explored the effect on response rates from different options to access an online survey from a mailed paper invitation: 1) a URL, 2) a QR code, and 3) a URL or QR code. For each of these conditions, the respondents were also given the option to complete the survey by filling the paper form (included with the invitation) and return it by mail. Marlar found no significant differences in response rates across the three conditions but found that only 4% of all respondents chose to access the survey via the QR code. The study also found that the condition that only included the QR code option showed a higher paper response than the other two conditions. Based on these findings, Marlar concluded that there was no value in including QR codes on survey invitations.

A couple of potential considerations from the Marlar (2018) study are worthy of note: First, data for this study were collected around 2017. At this point, QR codes were not as popular as they are today (Hamstra 2023, Ricson 2024). Second, the invitations in this study did not include any instructions beyond "Scan the QR code to access the website and complete the survey online". The lack of detailed instructions (about how to access the survey through the QR code), coupled with unfamiliarity with the process and, potentially, required third-party downloads, may have contributed to the low usage of the QR codes observed in the Marlar study.

As of today, the use of QR codes has become more prevalent compared to the years prior to the COVID-19 pandemic (Ricson 2024). According to Ceci (2023), the number of smartphone users who scanned a QR code in the United States in 2022 was 89 million (an increase of 26% compared to 2020). However, there is no recent research that has investigated the impact of QR codes on the survey-accessing experience, particularly if the QR code is accompanied by more detailed instructions. The purpose of the present study was two-fold: 1) To explore the usability of QR codes as a method to access an online survey on a smartphone, and 2) To determine the wording in the instructions that may allow novice respondents to successfully use a QR code. Specifically, we sought to answer the following questions:

To what extent does a QR code impact the respondent experience when accessing an online survey? and *What is the optimal wording of instructions that would allow non-tech-savvy individuals to use a QR code?*

To answer these questions, we investigated 1) the usability of QR codes as a method to access a mobile survey (in terms of effectiveness, efficiency, and satisfaction), as well as 2) respondents' opinions and suggestions for modifications to the instructions to ensure that this methodology could be used by anyone, regardless of QR code experience.

2. Methods

2.1 Participants

Twenty respondents aged 18-66 years (Mean age: 29.9 years; SD: 11.9 years; 11 females and 9 males) participated in this study between March and April 2022. All respondents were fluent

in English and were also iPhone users. The rationale for recruiting iPhone users was to ensure that the smartphone used for the study did not require a third-party application to read QR codes. Respondents were recruited through flyers and online postings in Texas and Pennsylvania, as well as through word-of-mouth. Usability sessions were conducted online through Microsoft Teams meetings. Each session was attended by one respondent, one researcher, and one note-taker. All respondents reported having obtained a high school diploma or higher. The sample's reported usage of a QR code during the year prior to the study is as follows: 40% weekly; 20% once every 2-3 weeks; 20% monthly use, and 20% once every 2-6 months. For level of comfort using a QR code, 95% of the sample reported being "very comfortable", and 5% of the sample reported being "comfortable". Prior to the session, all respondents provided written informed consent to participate in the study, and then were given a small honorarium upon completion of the session.

2.2 Design

The design of this study was observational. While respondents completed the study task, the researchers took notes on outcomes of interest (e.g., effectiveness, efficiency, satisfaction) and respondents' comments that may indicate a usability issue. Researchers also collected responses to debriefing questions and from feedback about the wording of the instructions to access the survey through a QR code.

2.3 Procedure

After joining the remote session through a desktop/laptop, respondents shared their screen so that the researchers could observe them completing the task. The test administrator sent the respondent a link via the chat functionality on the Teams software. The link took the respondent to a short (6-item) questionnaire. The specific wording of the questions can be found in Appendix A. Respondents followed the link and completed the questionnaire while thinking aloud. Towards the end of the questionnaire, respondents saw a screen with instructions to continue the survey on their smartphone. Figure 2 includes a screenshot of the instructions screen seen by the respondents.



Figure 2. Screen seen by respondents with instructions and QR code to continue the survey on their smartphone. Respondents were required to read aloud the text in this page.

After reading the instructions, and successfully scanning the QR code, respondents saw on their smartphone a screen that read "Please let the test administrator know that you have successfully opened the survey on your smartphone." The usability task concluded when users stated seeing this screen. After this, respondents completed three short debriefing questionnaires (satisfaction, technical difficulties and previous QR code experience questionnaires) and provided their general impression of the instructions to use the QR code.

2.4 Outcomes of interest

2.4.1 Quantitative outcomes

We measured effectiveness, efficiency, satisfaction, and technical difficulties while accessing the survey on a smartphone through a QR code. We also collected data about respondents' previous QR code experience.

Effectiveness was operationalized as the proportion of respondents who reported reading the statement "Please let the test administrator know that you have successfully opened the survey on your smartphone." This indicated that respondents had successfully scanned the QR code and were able to open the survey on their smartphone. This measure was recorded as a binary outcome (Read statement: Yes/No).

Efficiency was operationalized as the amount of time (in seconds) that it took the respondent to open the survey on their smartphone from the time they finished reading aloud the instructions in Figure 2, to the time that they read the statement presented in their smartphone.

Satisfaction was recorded by two different measures: Difficulty and likelihood of using a QR code in the future. *Difficulty* was operationalized as respondents' answer to the question "Overall, how easy or difficult was it to use the QR code to access the survey on your smartphone?" Response choices for this item included a 7-point response scale (1 = "Extremely easy" and 7 = "Extremely difficult"). *Likelihood of using a QR code in the future* was operationalized as respondents' answer to the question "If presented with the option, how likely are you to use a QR code to access a survey on your smartphone?" Response choices for this item included a 7-point response scale (1= "Extremely likely" and 7 = "Extremely unlikely").

Technical difficulties was measured by respondents' answer to two different questions. The first question asked the respondent "Did you have any technical difficulties scanning the QR code to access the survey on your smartphone?" The answer to this question was recorded as a binary outcome (yes/no). The second question included the wording "How many attempts did it take to successfully scan the QR code and access the survey on your smartphone?" and the numeric response given by the respondent was recorded.

Previous QR code experience was recorded as respondents' answer to four different questions: 1) Before today, did you know what a QR code was? 2) Before today, had you used QR codes? For these first two questions, response choices included "Yes" and "No"; 3) In the past year, on average, how frequently did you use QR codes? Response options for this question included: "Daily", "Weekly", "Once every 2-3 weeks", "Monthly", "Once every 2-6 months", and "Once a year"; and 4) In general, how comfortable are you using QR codes? Response choices for

this item included a 7-point response scale (1= “Very comfortable” and 7 = “Very uncomfortable”.)

2.4.2 Qualitative outcomes:

Besides the quantitative outcomes described above, researchers also recorded qualitative data in the form of notes about respondents’ behaviors and comments during the usability task and about the QR code process. Researchers also recorded respondents’ feedback about the instructions to use a QR code, and anecdotal accounts about respondents’ previous experience with QR codes (e.g., context where they first encountered QR codes.)

3. Data Analyses

3.1 Quantitative data:

Given the observational nature of the study, and the small sample size (n =20), quantitative data was summarized through descriptive statistics, and no inferential analyses were conducted.

3.2 Qualitative data:

Qualitative data recorded during the sessions was submitted to a content analysis (Stemler, 2015). The test administrator and the note-taker independently reviewed the notes recorded during the session. The researchers then categorized these notes into common themes and assigned labels depicting each category. After this, the researchers compared the categories’ labels, as well as the comments/actions included under each category and discussed the appropriateness of each label and categorization until agreement for the whole dataset was reached.

4. Results

4.1 Quantitative data

Previous experience using QR code. All respondents (100%) reported knowing what a QR code was and reported having used a QR code before the study. Most respondents (95% of sample) reported feeling “very comfortable” and 5% reported feeling “comfortable” using QR codes. Figure 3 shows respondents’ reported usage of QR codes during the year prior to the study.

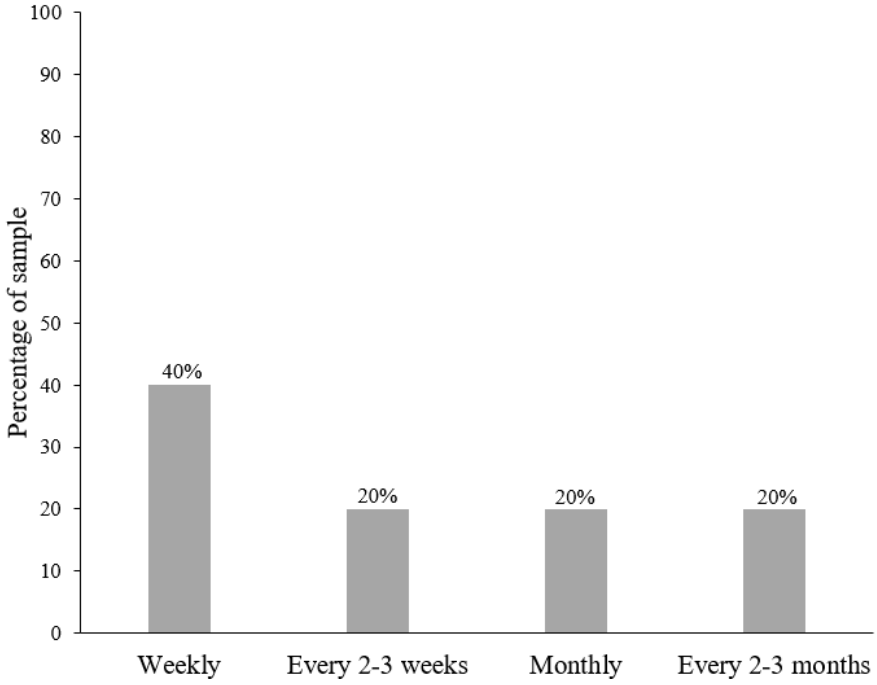


Figure 3. Frequency of responses to the question "In the past year, on average, how frequently did you use QR codes?" Note that almost half of the sample reported using this process on a weekly basis.

Effectiveness. All respondents (100%) were able to successfully scan the QR code to access the survey on their smartphone.

Efficiency. On average, respondents took 12.4 seconds (*SD* = 4.12) from the time they finished reading the instructions on their desktop/laptop, to the time they started reading the page in their phone which stated that they had successfully opened the survey on their smartphone.

Satisfaction. All respondents (100%) rated the process of using the QR code to access the survey on their smartphone as "extremely easy". Regarding the likelihood of using a QR code to access a survey (if presented with the option), most respondents (70%) reported being "extremely likely", and 20% of the sample reported being "likely" to use a QR code to access a survey on a smartphone. Two respondents (10% of the sample) reported being "somewhat unlikely" to access a survey through a smartphone. These respondents specified that this response was due to their preference for completing surveys on bigger screens (as opposed to a smartphone).

Technical difficulties. None of the respondents reported having any technical difficulties or needing help scanning the QR code to access the survey on their smartphone.

4.2 Qualitative data

Usability issues. No usability issues were observed during the process of respondents opening the survey on their smartphone through the QR code.

First experience using QR codes. When asked about their first experience using QR codes, 25% of respondents specifically remembered having to ask for help to complete the process. Most respondents (60%) remembered the context where they first learned to use a QR code. The rest of the sample (40%) reported learning this process so long ago that they didn't remember the exact context or situation. Figure 4 shows the distribution of responses for the places where respondents reported using a QR code for the first time.

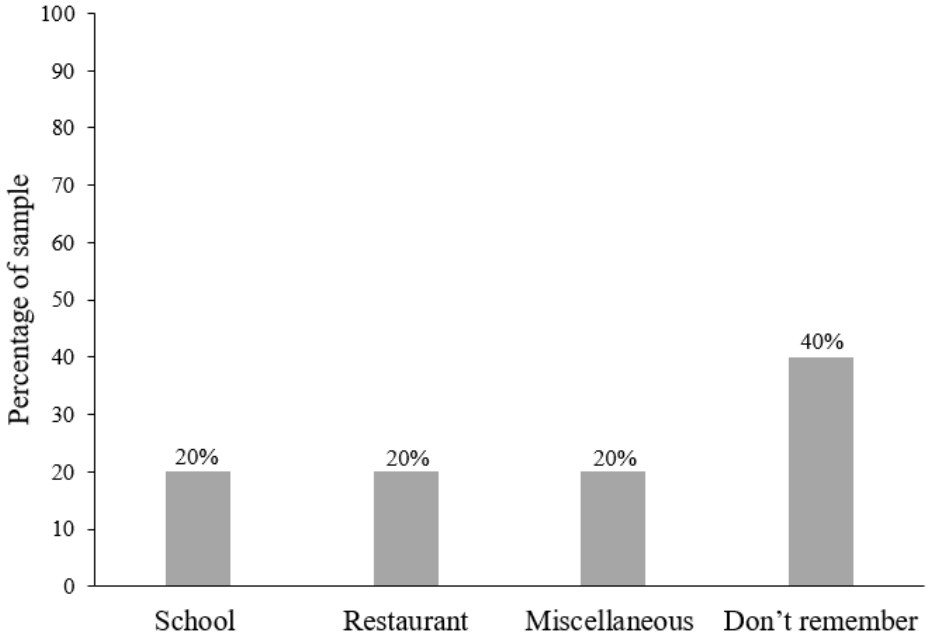


Figure 4. Frequency of responses about the context where respondents learned how to use a QR code. "Miscellaneous" category includes contexts that were only mentioned by a single respondent (e.g., work, online payment, etc.).

Impressions about instructions to use QR code. After being asked their general impression about the QR code instructions (instructions in Figure 2), respondents expressed that the instructions were "clear", "easy to follow/easy to understand", "straight-forward", and "simple". However, when respondents were asked if they thought someone without previous QR code experience would be able to follow the instructions, respondents expressed concern that "older people", "non-tech-savvy people" or "people for whom English is not their native language" may have trouble following the instructions. Specifically, respondents thought that these individuals would encounter at least one of the following problems:

- Thinking they need to use a different device (e.g., a camera device, as opposed to using the camera application on their phone).
- Believing they should take a picture of the QR code.
- Not understanding what "hover" means.
- Not understanding that the "code" is.
- Not knowing how to solve technical issues (e.g., link not opening because the QR code is not properly focused on camera).

- Not knowing that they must wait for the link to appear in their screen (thinking the process is not working because the link does not appear immediately).
- Not knowing how to "follow" the link (e.g., not knowing they must tap on the link after it appears on their screen).

When asked about their suggestions to solve the potential issues, respondents provided the following potential solutions:

- Specify "phone camera" (as opposed to "camera app").
- Replace "Hover" with "aim", "point", "scan", "hold camera over", "align" or "focus" so that respondents understand that they are not supposed to take a picture.
- Replace "code" with "square", "image" or "squared barcode".
- Label the QR code as "QR code".
- Outline the QR code with a color (e.g., red) and reference the color in the instructions (e.g., "scan the image in the red square").
- Specify that the code must be in focus.
- Specify that the link may not appear immediately.
- Specify that the link must be tapped to be opened.

Thoughts on using a QR code instead of typing a URL to access a survey on a mobile device. When asked about their general thoughts on the process of using a QR code to access a mobile survey, all respondents expressed excitement and used positive words to describe why they thought this process was better than typing a URL. Figure 5 shows the frequency of words/concepts expressed by respondents to describe why a QR code would be a preferred method to access a mobile survey. Two respondents, however, specified the caveat that typing a short URL (e.g., somepage.com) would be tolerable, but they would prefer to scan a QR code if the URL was longer.

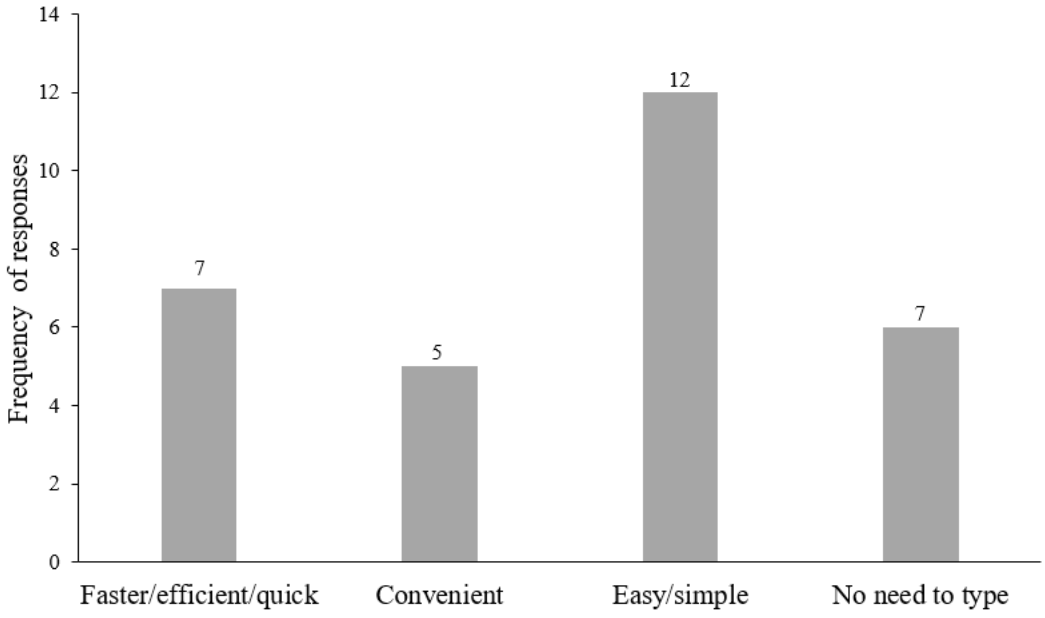


Figure 5. Frequency of concepts that respondents spontaneously used to describe the QR code process to access a survey. *Note*: Because some respondents expressed more than one concept, the total frequency of responses is larger than the sample size.

5. Discussion

In this study, we developed step-by-step instructions to guide potential respondents to access a mobile survey using a QR code, and sought to answer the questions "*To what extent does a QR code impact the respondent experience when accessing an online survey"? and "What is the optimal wording of instructions that would allow non-tech-savvy individuals to use a QR code?"*

Regarding the respondents' experience, the results of this study indicated that QR codes have high usability as a methodology to access a survey: All respondents were able to access the survey using the QR code, within a short time, and rated the process as extremely easy. In fact, most respondents reported that, if presented with a QR code option to complete the survey, they would be extremely likely to use this method to complete the survey instead of choosing to type a URL. Here, it is worthy of note that the high usability of the QR code process may be due to our sample's previous experience with this process, and not to the clarity of the instructions. Some respondents spontaneously commented "I knew what to do as soon as I saw the QR code," "I knew I needed my phone when I saw the QR code," and "I could have done this without reading the instructions."

Although we aimed to recruit respondents without QR code experience to tailor the wording of the instructions, we were unable to find respondents with this characteristic. Here, it is important to note that the recruitment material did not include any information about QR codes, precisely to avoid discouraging novice respondents from participating. Still, with this measure in place, we were unable to recruit respondents without previous QR code experience. Because of this, the recommendations to improve our instructions were based on what experienced respondents thought would be helpful for novice respondents. The familiarity of our sample with the QR code process can be interpreted as a limitation of this study. However, this familiarity may be good news for organizations seeking to provide QR codes as an option to sign into electronic surveys.

During debriefing, most respondents made spontaneous comments about how QR codes are more prevalent in everyday activities since the COVID-19 pandemic. In fact, most respondents specifically stated learning how to use a QR code during 2020. One respondent also stated that "…these days, not knowing how to use a QR code is like not knowing how to hail a cab." However, when asked about their first experience with QR codes, some respondents specifically recalled having to ask for help, but not having any issues after using QR codes for the first time.

The statements above suggest that, although using a QR code may not be an intuitive process, it is relatively easy-to-learn, and it has become a common process in the last few years. The sample's spontaneous comments about the prevalence of QR codes after the COVID-19 pandemic echoes the findings from technology and marketing companies who have reported a substantial increase in generation and usage of QR codes in the past few years (Ricson 2024; Ceci 2023). In fact, the number of smartphone users in the United States who scan a QR code is predicted to reach 100 million by 2025 (Ceci 2023, Hamstra 2023).

6. Conclusions and Future directions

Although six years ago Marlar (2018) concluded that the benefits of including a QR code to access a mobile survey did not justify the cost associated with it, the results of our study allow a different conclusion: Given the high usability of the QR code process observed in this study, the sample's self-reported likelihood of using this method to access a survey, and our inability to find respondents without QR code experience, we conclude that providing this option may increase the respondents' satisfaction with said survey, reduce respondents' burden, and potentially increase survey response. Furthermore, given that most of our respondents reported being highly likely to use QR codes instead of typing a website address with strings of characters, providing a QR code option may be particularly beneficial for surveys that require respondents to enter a long, personalized passcode to log into the survey.

Given our inability to recruit novice respondents, future studies should investigate whether a sample without previous QR code experience would be able to follow the instructions with the modifications suggested by the expert sample in this study. Furthermore, given the increased prevalence of QR Codes since Marlar (2018), a replication of their study is due, but with a sample that represents the population's current experience with the QR Code process.

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Appendix A

Demographic questionnaire

1. As of today, how old are you? [open ended; numeric]

2. What is the highest level of education you have completed?
    - Some high school, no diploma
    - High school graduate or the equivalent (for example, GED)
    - Some college credit, no degree
    - Trade/technical/vocational training
    - Associates degree
    - Bachelor's degree
    - Professional Degree
    - Doctorate degree

3. What is your gender? [open ended]

4. What is your race? Please select all that apply
    - White
    - Black or African American
    - American Indian or Alaska Native
    - Asian
    - Native Hawaiian or Other Pacific Islander

5. Are you of Hispanic, Latino, of Spanish origin? [yes/no]

6. What model of iPhone are you currently using? (e.g., iPhone 6, iPhone 12 mini). [open ended]